



East West University

Department of Computer Science and Engineering

Project Description [Fall 2025]

E-Learning Course Management System

1.Introduction:

E-learning platforms are becoming an essential part of modern education systems. Managing courses, students, and enrollments manually can be time-consuming and error-prone. To overcome these challenges, a computer-based solution is required that can efficiently store, manage, and retrieve academic information. This project, **E-Learning Course Management System**, is developed using the **C programming language**. It is a console-based, menu-driven application that allows users to manage courses, student records, and course enrollments effectively. The system demonstrates the use of structures, arrays, functions, loops, and basic data management techniques in C.

2. Project Overview:

The E-Learning Course Management System helps manage academic data in an organized manner. It allows users to:

- Add and view courses
- Add and view students
- Enroll students into courses
- Display enrollment information

The project demonstrates the practical use of fundamental C programming concepts such as **structures, arrays, functions, loops, and conditional statements** to build a simple, interactive management system.

Managing course and student information manually can lead to:

- Data duplication and inconsistency
- Difficulty in tracking student enrollments
- Inefficient record keeping
- Higher chances of human error

The problem is to design a system that can:

- Store course and student data securely
- Retrieve and display information easily
- Assign students to courses efficiently

3. Code Explanation:

The program is developed using basic C programming concepts:

- **Structures** store course, student, and enrollment information. They organize related data into a single unit, making the program easy to manage.

- **Arrays** hold multiple records during execution, allowing efficient handling of many data entries.
- **Menu-driven system** provides an interactive interface for the user to choose different operations.
- **Functions** perform specific tasks like adding courses, adding students, enrolling students, and displaying data. They make the program modular, reusable, and easy to debug.
- **Conditional statements and loops** control program flow, while ID-based checks ensure only valid operations are performed.

4. Solutions:

- **Modular Design:** The program divides the tasks into multiple functions like adding courses, adding students, enrolling students, and displaying records.
- **Data Storage:** Uses arrays of structures to store course, student, and enrollment data.
- **Validation:** Checks whether a course or student exists before enrolling to avoid errors.
- **User-Friendly Interface:** Uses menus and headings to guide users clearly through available options.
- **Scalable:** Can easily increase the number of courses and students by adjusting the MAX size.
- **Console-based Interaction:** Ensures simplicity for demonstration and testing purposes.

5. Sample Input Output (Screenshots):

Menu Options:

- Add Course

- List Courses
- Add Student
- List Students
- Enroll Student
- View Enrollments
- Exit

List of the functions that should be used in the “E-Learning Course Management System”

Function Descriptions:

- **Welcome_Message():**

```
=====
||                               ||
||      E-LEARNING COURSE MANAGEMENT SYSTEM      ||
||                               ||
=====
||                               ||
|| Welcome to the E-Learning Management System project! ||
||                               ||
|| -> Developed by: Department of CSE                ||
|| -> Semester   : Fall 2025                        ||
||                               ||
|| This system allows you to:                       ||
||   - Add and manage courses                       ||
||   - Add and manage students                      ||
||   - Enroll students into courses                 ||
||   - View enrollment information                   ||
||                               ||
=====
||                               ||
|| Press any key to start the system...             ||
||                               ||
=====
```

- **Menu():**

```
-----  
Main Menu  
-----  
1. Add Course  
2. List Courses  
3. Add Student  
4. List Students  
5. Enroll Student  
6. View Enrollments  
0. Exit
```

- **Add_Courses():**

```
-----  
Main Menu  
-----  
1. Add Course  
2. List Courses  
3. Add Student  
4. List Students  
5. Enroll Student  
6. View Enrollments  
0. Exit  
Enter your choice: 1  
  
-----  
Add Course  
-----  
Enter Course ID: 101  
Enter Course Name: CSE  
Course Added Successfully!
```

```
-----  
Main Menu  
-----  
1. Add Course  
2. List Courses  
3. Add Student  
4. List Students  
5. Enroll Student  
6. View Enrollments  
0. Exit  
Enter your choice: 1  
  
-----  
Add Course  
-----  
Enter Course ID: 102  
Enter Course Name: CSE  
Course Added Successfully!
```

• **List_Courses():**

```
-----  
Main Menu  
-----  
1. Add Course  
2. List Courses  
3. Add Student  
4. List Students  
5. Enroll Student  
6. View Enrollments  
0. Exit  
Enter your choice: 2  
  
-----  
Course List  
-----  
ID: 101 | Name: CSE  
ID: 102 | Name: CSE
```

- **Add_Student():**

```
-----  
Main Menu  
-----  
1. Add Course  
2. List Courses  
3. Add Student  
4. List Students  
5. Enroll Student  
6. View Enrollments  
0. Exit  
Enter your choice: 3  
  
-----  
Add Student  
-----  
Enter Student ID: 1000  
Enter Student Name: Hema  
Student Added Successfully!
```

```
-----  
Main Menu  
-----  
1. Add Course  
2. List Courses  
3. Add Student  
4. List Students  
5. Enroll Student  
6. View Enrollments  
0. Exit  
Enter your choice: 3  
  
-----  
Add Student  
-----  
Enter Student ID: 2000  
Enter Student Name: Shayori  
Student Added Successfully!
```

- **List_Students():**

```
-----  
Main Menu  
-----  
1. Add Course  
2. List Courses  
3. Add Student  
4. List Students  
5. Enroll Student  
6. View Enrollments  
0. Exit  
Enter your choice: 4  
  
-----  
Student List  
-----  
ID: 1000 | Name: Hema  
ID: 2000 | Name: Shayori
```

- **Enroll_Student():**

```
-----  
Main Menu  
-----  
1. Add Course  
2. List Courses  
3. Add Student  
4. List Students  
5. Enroll Student  
6. View Enrollments  
0. Exit  
Enter your choice: 5  
  
-----  
Enroll Student  
-----  
Enter Student ID: 1000  
Enter Course ID: 101  
Enrollment Successful!
```

```
-----  
Main Menu  
-----  
1. Add Course  
2. List Courses  
3. Add Student  
4. List Students  
5. Enroll Student  
6. View Enrollments  
0. Exit  
Enter your choice: 5  
  
-----  
Enroll Student  
-----  
Enter Student ID: 2000  
Enter Course ID: 102  
Enrollment Successful!
```

- **View_Enrollments():**

```
-----  
Main Menu  
-----  
1. Add Course  
2. List Courses  
3. Add Student  
4. List Students  
5. Enroll Student  
6. View Enrollments  
0. Exit  
Enter your choice: 6  
  
-----  
Enrollment List  
-----  
Student ID: 1000 --> Course ID: 101  
Student ID: 2000 --> Course ID: 102
```

- **Exit_Program():**

Displays a thank you message and ends the program.

```
-----  
                          Main Menu  
-----  
1. Add Course  
2. List Courses  
3. Add Student  
4. List Students  
5. Enroll Student  
6. View Enrollments  
0. Exit  
Enter your choice: 0  
  
-----  
                          Exit  
-----  
  
=====
```

Thank you for using E-Learning System!

```
=====
```

6.Conclusion:

The **E-Learning Course Management System** is a simple console-based application developed in C. It effectively demonstrates the use of structures, arrays, and functions to manage courses and students. The system is user-friendly and provides a strong foundation in data management and programming concepts. While it has limitations like temporary data storage, it can be further enhanced with features like file handling and authentication. Overall, the project strengthens problem-solving skills and practical C programming knowledge.